

SQL Practice Task – Subqueries (Part 1)

Topics Covered

- Scalar Subqueries
- Multi-row Subqueries
- Correlated Subqueries

Instructions

- Create the database and tables as instructed.
 - Insert your own sample data (minimum 10 records in each table unless specified).
 - Solve every question **using only Subqueries**.
 - **Do NOT use JOINS.**
 - Execute every query and verify the output.
-

Database Scenario

You are developing a database for a company called **TechNova Solutions**.

The company has two tables:

- Departments
- Employees

Employees belong to different departments, and the management wants various analytical reports using **Subqueries**.

Create the Tables

Create the following tables:

departments

Column	Data Type
id	INT
department_name	VARCHAR(50)
location	VARCHAR(50)

```
create table departments(
```

```
id int primary key,
```

```
department_name varchar(50),
```

```
location varchar(50));  
  
insert into departments values  
(1,'Engineering','Hyderabad'),  
(2,'Sales','Bangalore'),  
(3,'HR','Hyderabad'),  
(4,'Finance','Chennai')  
;
```

employees

Column	Data Type
id	INT
name	VARCHAR(50)
salary	DECIMAL(10,2)
department_id	INT

Use appropriate **PRIMARY KEY** and **FOREIGN KEY** constraints.

Insert:

- At least **4 Departments**
- At least **10 Employees**

Make sure:

- Multiple employees belong to the same department.
- Different departments have different average salaries.
- At least two departments are in the same city.

```
create table employees(  
id int primary key,  
name varchar(50),  
salary decimal(10,2),  
department_id int,  
foreign key(department_id) references departments(id)  
);  
  
insert into employees values  
(101,'Dinesh',80000,1),
```

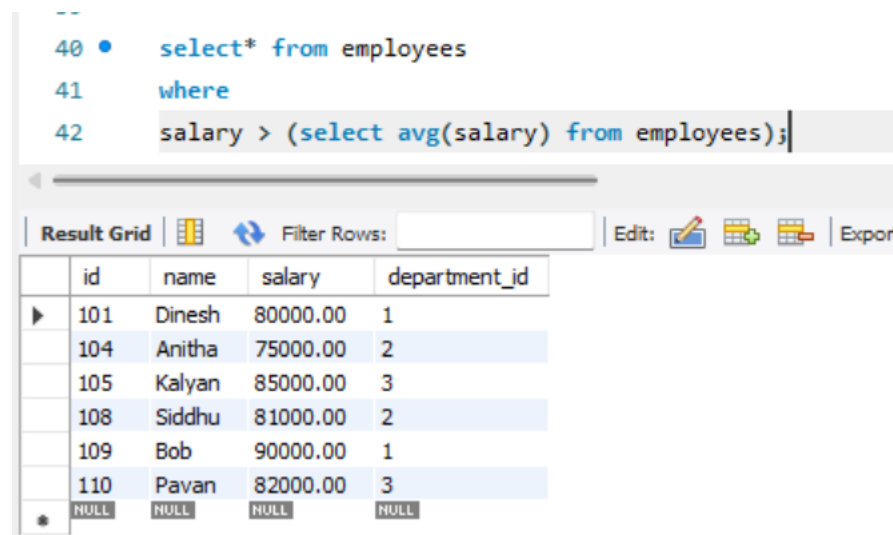
(102,'Akhila',68000,2),
(103,'Sushma',65000,1),
(104,'Anitha',75000,2),
(105,'Kalyan',85000,3),
(106,'Babu',35000,4),
(107,'Vignesh',40000,4),
(108,'Siddhu',81000,2),
(109,'Bob',90000,1),
(110,'Pavan',82000,3)
;

Section A – Scalar Subqueries

Use **only Scalar Subqueries**.

1.

Display employees earning more than the company average salary.



```
40 • select* from employees
41 where
42 salary > (select avg(salary) from employees);
```

Result Grid

	id	name	salary	department_id
▶	101	Dinesh	80000.00	1
	104	Anitha	75000.00	2
	105	Kalyan	85000.00	3
	108	Siddhu	81000.00	2
	109	Bob	90000.00	1
	110	Pavan	82000.00	3
*	NULL	NULL	NULL	NULL

2.

Display the employee(s) earning the highest salary.

```
44 • select * from employees
45 where
46 salary = (select max(salary) from employees);
```

Result Grid					Filter Rows:		Edit:			Export
	id	name	salary	department_id						
▶	109	Bob	90000.00	1						
*	NULL	NULL	NULL	NULL						

3.

Display the employee(s) earning the lowest salary.

```
48 • select * from employees
49 where
50 salary = (select min(salary) from employees);
```

Result Grid					Filter Rows:		Edit:			Export
	id	name	salary	department_id						
▶	106	Babu	35000.00	4						
*	NULL	NULL	NULL	NULL						

4.

Display employees earning more than the minimum salary.

```

52 • select * from employees
53 where
54 salary > (select min(salary) from employees);

```

Result Grid		Filter Rows:		Edit:			Exp
	id	name	salary	department_id			
▶	101	Dinesh	80000.00	1			
	102	Akhila	68000.00	2			
	103	Sushma	65000.00	1			
	104	Anitha	75000.00	2			
	105	Kalyan	85000.00	3			
	107	Vignesh	40000.00	4			
	108	Siddhu	81000.00	2			
	109	Bob	90000.00	1			
	110	Pavan	82000.00	3			
*	NULL	NULL	NULL	NULL			

5.

Display employees earning less than the highest salary.

```

56 • select * from employees
57 where
58 salary < (select max(salary) from employees);

```

Result Grid		Filter Rows:		Edit:			
	id	name	salary	department_id			
▶	101	Dinesh	80000.00	1			
	102	Akhila	68000.00	2			
	103	Sushma	65000.00	1			
	104	Anitha	75000.00	2			
	105	Kalyan	85000.00	3			
	106	Babu	35000.00	4			
	107	Vignesh	40000.00	4			
	108	Siddhu	81000.00	2			
	110	Pavan	82000.00	3			
*	NULL	NULL	NULL	NULL			

6.

Display employees earning exactly the average salary.

Observe the output.

```

60 • select * from employees
61 where
62 salary=(select avg(salary) from employees);

```

Result Grid

Filter Rows:

Edit:

	id	name	salary	department_id
*	NULL	NULL	NULL	NULL

7.

Display employees earning greater than the average salary but less than the highest salary.

```

64 • select * from employees
65 where
66 salary > (select avg(salary) from employees) and salary < (select max(salary) from employees);

```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	id	name	salary	department_id
▶	101	Dinesh	80000.00	1
	104	Anitha	75000.00	2
	105	Kalyan	85000.00	3
	108	Siddhu	81000.00	2
	110	Pavan	82000.00	3
*	NULL	NULL	NULL	NULL

8.

Display employees earning less than the company average but greater than the minimum salary.

```

68 • select * from employees
69 where
70 salary < (select avg(salary) from employees) and salary > (select min(salary) from employees);

```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	id	name	salary	department_id
▶	102	Akhila	68000.00	2
	103	Sushma	65000.00	1
	107	Vignesh	40000.00	4
*	NULL	NULL	NULL	NULL

9.

Find employees earning exactly the maximum salary.

```
72 • select * from employees
73 where
74 salary=(select max(salary) from employees);
```

Result Grid | Filter Rows: | Edit: | Export/Import

	id	name	salary	department_id
▶	109	Bob	90000.00	1
*	NULL	NULL	NULL	NULL

10.

Find employees earning exactly the minimum salary.

```
76 • select * from employees
77 where
78 salary=(select min(salary) from employees);
```

Result Grid | Filter Rows: | Edit: | Export/Import

	id	name	salary	department_id
▶	106	Babu	35000.00	4
*	NULL	NULL	NULL	NULL

Section B – Multi-row Subqueries

Use **only Multi-row Subqueries** with the **IN** operator.

11.

Display employees working in Hyderabad departments.

```

80 • select * from employees
81 where
82 department_id in (select id from departments where location='Hyderabad');

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Cont
	id	name	salary	department_id
▶	101	Dinesh	80000.00	1
	103	Sushma	65000.00	1
	109	Bob	90000.00	1
	105	Kalyan	85000.00	3
	110	Pavan	82000.00	3
*	NULL	NULL	NULL	NULL

12.

Display employees working in Bangalore departments.

```

84 • select * from employees
85 where
86 department_id in (select id from departments where location = 'Bangalore');

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Cont
	id	name	salary	department_id
▶	102	Akhila	68000.00	2
	104	Anitha	75000.00	2
	108	Siddhu	81000.00	2
*	NULL	NULL	NULL	NULL

13.

Display employees working in Chennai departments.

```

88 • select * from employees
89 where
90 department_id = (select id from departments where location = 'Chennai');

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell
	id	name	salary	department_id
▶	106	Babu	35000.00	4
	107	Vignesh	40000.00	4
*	NULL	NULL	NULL	NULL

14.

Display employees working in Engineering or Sales departments.

```
92 • select * from employees
93   where
94   department_id in (select id from departments
95   where
96   department_name='Engineering' or department_name='Sales');
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	id	name	salary	department_id
▶	101	Dinesh	80000.00	1
	103	Sushma	65000.00	1
	109	Bob	90000.00	1
	102	Akhila	68000.00	2
	104	Anitha	75000.00	2
	108	Siddhu	81000.00	2
*	NULL	NULL	NULL	NULL

15.

Display employees belonging to departments whose ID is greater than 2.

```
98 • select * from employees
99   where
100   department_id in (select id from departments where id > 2);
101
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	id	name	salary	department_id
▶	105	Kalyan	85000.00	3
	110	Pavan	82000.00	3
	106	Babu	35000.00	4
	107	Vignesh	40000.00	4
*	NULL	NULL	NULL	NULL

16.

Display employees belonging to departments located in Hyderabad or Chennai.

```
104 • select * from employees
105 where
106 department_id in (select id from departments
107 where location = 'Hyderabad' or location='Chennai');
```

Result Grid

	id	name	salary	department_id
▶	101	Dinesh	80000.00	1
	103	Sushma	65000.00	1
	109	Bob	90000.00	1
	105	Kalyan	85000.00	3
	110	Pavan	82000.00	3
	106	Babu	35000.00	4
	107	Vignesh	40000.00	4
*	NULL	NULL	NULL	NULL

17.

Display employees working in departments whose names start with the letter **S**.

```
109 • select * from employees
110 where
111 department_id in (select id from departments where department_name like 's%');
```

Result Grid

	id	name	salary	department_id
▶	102	Akhila	68000.00	2
	104	Anitha	75000.00	2
	108	Siddhu	81000.00	2
*	NULL	NULL	NULL	NULL

18.

Display employees working in departments located outside Bangalore.

```

113 • select * from employees
114 where
115 department_id in (select id from departments where
116 location != 'Bangalore');

```

Result Grid	Filter Rows:	Edit:	Export/Im
id	name	salary	department_id
101	Dinesh	80000.00	1
103	Sushma	65000.00	1
109	Bob	90000.00	1
105	Kalyan	85000.00	3
110	Pavan	82000.00	3
106	Babu	35000.00	4
107	Vignesh	40000.00	4
NULL	NULL	NULL	NULL

19.

Display employees belonging to departments with IDs less than 3.

```

118 • select * from employees
119 where
120 department_id in (select id from departments where id < 3);

```

Result Grid	Filter Rows:	Edit:	Export/Import:
id	name	salary	department_id
101	Dinesh	80000.00	1
103	Sushma	65000.00	1
109	Bob	90000.00	1
102	Akhila	68000.00	2
104	Anitha	75000.00	2
108	Siddhu	81000.00	2
NULL	NULL	NULL	NULL

20.

Display employees belonging to all departments located in Hyderabad.

```

122 • select * from employees
123 where
124 department_id in (select id from departments where location = 'Hyderabad');

```

Result Grid Filter Rows: | Edit: | Export/Import: | Wrap Cell Conte

	id	name	salary	department_id
▶	101	Dinesh	80000.00	1
	103	Sushma	65000.00	1
	109	Bob	90000.00	1
	105	Kalyan	85000.00	3
	110	Pavan	82000.00	3
*	NULL	NULL	NULL	NULL

Challenge Question

Try replacing **IN** with **=** in one of the above queries.

Observe the error and explain why it occurs.

```

select * from employees where department_id = (select id from departments where location = 'Hyderabad') LIMIT 0, 1000
0 19 17:49:35 Error Code: 1242. Subquery returns more than 1 row 0.000 sec

```

Section C – Correlated Subqueries

Use **only Correlated Subqueries**.

21.

Display employees earning more than the average salary of their own department.

```

126 • select * from employees e
127     where
128     salary > (select avg(salary) from employees
129     where department_id = e.department_id);

```

Result Grid	Filter Rows:	Edit:	Export/
id	name	salary	department_id
101	Dinesh	80000.00	1
104	Anitha	75000.00	2
105	Kalyan	85000.00	3
107	Vignesh	40000.00	4
108	Siddhu	81000.00	2
109	Bob	90000.00	1
NULL	NULL	NULL	NULL

22.

Display the highest paid employee from each department.

```

131 • select * from employees e
132     where
133     salary = (select max(salary) from employees where department_id = e.department_id);

```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	id	name	salary	department_id
▶	105	Kalyan	85000.00	3
	107	Vignesh	40000.00	4
	108	Siddhu	81000.00	2
	109	Bob	90000.00	1
*	NULL	NULL	NULL	NULL

23.

Display the lowest paid employee from each department.

```

135 • select * from employees e
136     where
137     salary =(select min(salary) from employees where department_id=e.department_id);

```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	id	name	salary	department_id
▶	102	Akhila	68000.00	2
	103	Sushma	65000.00	1
	106	Babu	35000.00	4
	110	Pavan	82000.00	3
✱	NULL	NULL	NULL	NULL

24.

Display employees earning exactly the average salary of their department.

Observe the output.

```
139 • select * from employees e
140 where
141 salary = (select avg(salary) from employees where department_id=e.department_id);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	id	name	salary	department_id
*	NULL	NULL	NULL	NULL

25.

Display employees earning above their department average but below the company average.

```
143 • select * from employees e
144 where
145 salary > (select avg(salary) from employees where department_id=e.department_id)
146 and
147 salary < (select avg(salary) from employees);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	id	name	salary	department_id
▶	107	Vignesh	40000.00	4
*	NULL	NULL	NULL	NULL

26.

Display employees earning below their department average.

```

149 • select * from employees e
150 where
151 salary < (select avg(salary) from employees where department_id=e.department_id);

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
id	name	salary	department_id	
102	Akhila	68000.00	2	
103	Sushma	65000.00	1	
106	Babu	35000.00	4	
110	Pavan	82000.00	3	
NULL	NULL	NULL	NULL	

27.

Display employees whose salary is equal to the maximum salary in their department.

```

154 • select * from employees e
155 where
156 salary = (select max(salary) from employees where department_id=e.department_id);

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
id	name	salary	department_id	
105	Kalyan	85000.00	3	
107	Vignesh	40000.00	4	
108	Siddhu	81000.00	2	
109	Bob	90000.00	1	
NULL	NULL	NULL	NULL	

28.

Display employees whose salary is equal to the minimum salary in their department.

```

158 • select * from employees e
159 where
160 salary = (select min(salary) from employees where department_id = e.department_id);

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
id	name	salary	department_id	
102	Akhila	68000.00	2	
103	Sushma	65000.00	1	
106	Babu	35000.00	4	
110	Pavan	82000.00	3	
NULL	NULL	NULL	NULL	

29.

Display employees whose salary is greater than every other employee in their own department.

```
162 • select * from employees e
163 where
164 salary = (select max(salary) from employees where department_id = e.department_id);
165
```

Result Grid				
Filter Rows:				
Edit: Export/Import: Wrap Cell Content:				
	id	name	salary	department_id
▶	105	Kalyan	85000.00	3
	107	Vignesh	40000.00	4
	108	Siddhu	81000.00	2
	109	Bob	90000.00	1
*	NULL	NULL	NULL	NULL

30.

Display employees whose salary is less than every other employee in their own department.

```
166 • select * from employees e
167 where
168 salary = (select min(salary) from employees where department_id = e.department_id);
169
```

Result Grid				
Filter Rows:				
Edit: Export/Import: Wrap Cell Content:				
	id	name	salary	department_id
▶	102	Akhila	68000.00	2
	103	Sushma	65000.00	1
	106	Babu	35000.00	4
	110	Pavan	82000.00	3
*	NULL	NULL	NULL	NULL